



AI-Based Ecological Predictive Cruise Control for Connected EVs

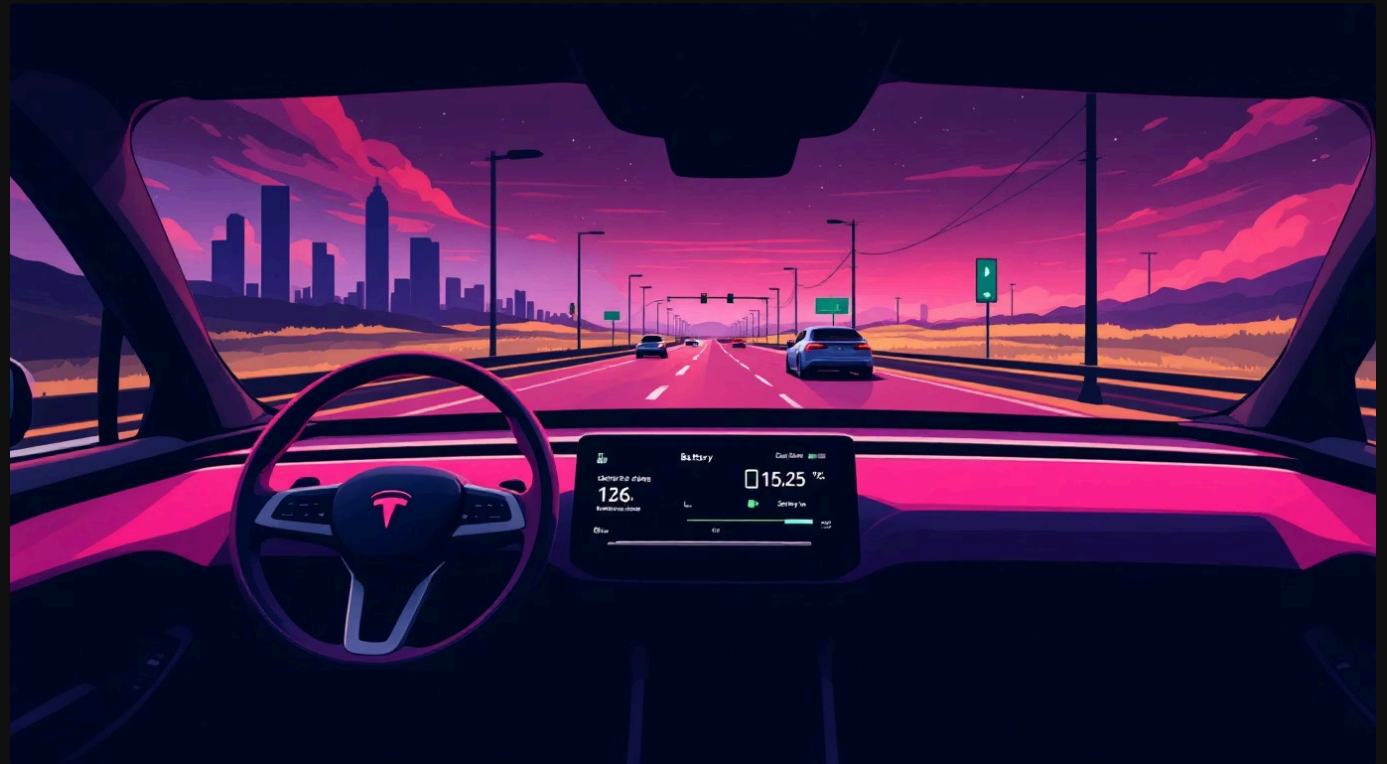
Optimizing Energy Efficiency in Connected Electric Vehicles through Deep Learning and Model Predictive Control

The Core Problem

Current Limitations

Standard Adaptive Cruise Control (ACC) operates purely reactively, responding to immediate changes in traffic conditions. This results in delayed braking decisions, harsh deceleration events, and significant energy waste through inefficient regenerative braking cycles.

Traditional PID controllers lack foresight, treating each moment independently rather than anticipating future traffic patterns. The result: erratic speed profiles that drain battery capacity unnecessarily.



Our Solution Framework



LSTM Predictor

Deep neural network forecasts leader vehicle behavior 10+ seconds ahead using traffic pattern analysis



MPC Controller

Model Predictive Control optimizes acceleration and braking commands in real-time using rolling horizon optimization



EV Digital Twin

High-fidelity physics model simulates aerodynamic drag, rolling resistance, and asymmetric powertrain efficiency

This integrated approach transforms cruise control from a reactive system into an **intelligent eco-driving assistant** that maximizes regenerative braking opportunities while maintaining optimal following distances.



Technical Architecture

Deep LSTM Network

Long Short-Term Memory architecture with dropout and batch normalization trained on multi-scenario synthetic traffic data

Multi-Objective Cost Function

Weighted optimization balancing target tracking, safety margins, energy consumption, and ride comfort metrics

SLSQP Solver

Sequential Least Squares Programming (SciPy) executes 10-step receding horizon, updated every 1 second

Intelligent Traffic Prediction

LSTM Architecture

- Multiple hidden layers with dropout regularization
- Batch normalization for training stability
- 20-step temporal horizon prediction

The network learns chaotic traffic patterns from synthetic scenarios, enabling **proactive speed adjustments** before disturbances occur.



Energy Optimization Model

92%

Motor
Efficiency

Forward driving
efficiency under
acceleration

75%

Regen
Efficiency

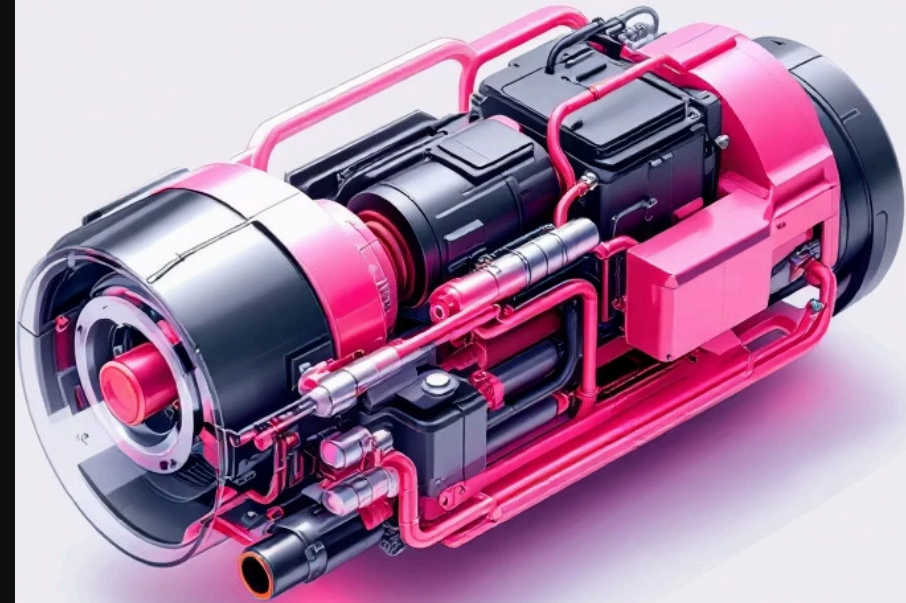
Energy recovery during
braking events

~13%

Energy Savings

Compared to
traditional PID
controllers

The digital twin accurately models asymmetric powertrain characteristics, enabling MPC to maximize regenerative braking while minimizing resistive losses from aerodynamic drag and rolling resistance.



Model Predictive Control Strategy

01

Predict Traffic State

LSTM forecasts leader vehicle speed profile over 20-step horizon (10 seconds)

02

Optimize Torque Commands

SLSQP solver minimizes cost function balancing safety, energy, and comfort weights

03

Execute First Command

Apply optimal motor torque to ego vehicle while re-planning next iteration

04

Update & Repeat

Continuous feedback loop adjusts predictions based on actual traffic conditions



Performance Comparison

Traditional ACC

- Reactive braking responses
- High jerk values
- Inefficient regen utilization
- ~13% more energy consumed

Eco-PCC

- Predictive speed management
- Smooth acceleration profiles
- Maximized regenerative braking
- Up to ~13% battery savings

Key Technical Achievements



Enhanced Range

Up to ~13% reduction in energy consumption through predictive coasting and regenerative braking optimization



Safety First

Dynamic following distance maintenance with multi-objective cost function prioritizing collision avoidance



Superior Comfort

Minimized jerk values through smooth acceleration profiles, eliminating harsh braking events

By combining deep learning with optimal control theory, Eco-PCC transforms cruise control into an **intelligent eco-driving system** that extends EV range without requiring larger batteries.

References & Research Foundation

Foundational Research

Eco-Driving Strategy for Electric Vehicles Based on Machine Learning and MPC

LSTM Architecture

Hochreiter, S., & Schmidhuber, J. (1997). Long Short-Term Memory. Neural Computation

Optimization Methods

SciPy Documentation: SLSQP Sequential Least Squares Programming Solver

Deep Learning Tools

TensorFlow/Keras Documentation for Neural Network Implementation

This project leverages industry-standard libraries and established control theory to create an implementable solution for production electric vehicles.